SXFunction_constr_py

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This file is part of CasADi.

CasADi -- A symbolic framework for dynamic optimization.
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1 Function constructors

[1]:
```
import casadi as ca
```

[2]:
```
x = ca.SX.sym("x")      # A scalar (1-by-1 matrix) symbolic primitive
y = ca.SX.sym("y",2)    # A vector (n-by-1 matrix) symbolic primitive
z = ca.SX.sym("z",2,3) # An n-by-m matrix symbolic primitive
```

[3]:
```
ins =  [x,y] # function inputs
outs = [x,y,ca.vertcat(x,y),y*x,0]
```

[4]:
```
print(outs)
```

```
[SX(x), SX([y_0, y_1]), SX([x, y_0, y_1]), SX([(y_0*x), (y_1*x)]), 0]
```

[5]:
```
f = ca.Function("f", ins, outs)
```

f now has two inputs and a 4 outputs:

[6]:
```
print(f.n_in())
print(f.n_out())
```

```
2
5
```

The outputs has the following string representation. Note how all elements of out have been converted to SX by automatic typecasting functionality

[7]:
```
f_out = f(*f.sx_in())
for i in range(3):
  print(f_out[i])
```

```
x
[y_0, y_1]
[x, y_0, y_1]
```